1. Explain Binomial, Poisson, and Normal distributions. Point out their important properties and applications

ANSWER

Binomial Probability Distribution

Let there be n independent trials in an experiment. Let a random variable X denote the number of success in these n trials. Let p be the probability of a success and q that of a failure in a single trial so that $p+q=1$. Let the trials be independent and p be constant for every trial.

Let us find the probability of r successful in n trials. r successes can be obtained in n trials in ^nC_r ways.

$$\therefore P(X=x) = {}^nC_x\, P(\underbrace{SSS\ldots S}_{x \text{ times.}}\ \underbrace{FFF\ldots F}_{(n-x) \text{ times.}})$$

$$= {}^nC_x\, \underbrace{P(S)\,P(S)\ldots P(S)}_{x \text{ factors.}}\ \underbrace{P(F).P(F)\ldots P(F)}_{(n-x) \text{ factors.}}$$

$$= {}^nC_x\, \underbrace{p\,p\,p\ldots p}_{x \text{ factors}}\ \underbrace{q\,q\,q\ldots q}_{(n-x) \text{ factors.}}$$

$$= {}^nC_x\, p^x q^{n-x}.$$

Hence $P(X=x) = {}^nC_x\, q^{n-x} p^x$; where $p+q=1$ & $x=0,1,2,\ldots n$ —(1)

The distribution (1) is called binomial distribution and X is called binomial variate.

NOTE 1 ÷ ~~$P(x=x) = {}^nC_x q^{n-x} p^x$~~

$P(x=r)$ is usually written as $P(r)$.

NOTE 2 ÷ The successive probabilities $P(r)$ in (1) for $r = 0, 1, 2, \ldots n$

are $nC_0 q^n$, $nC_1 q^{n-1}$, $nC_2 q^{n-2}$ .... $nC_n p^n$.

which are the successive terms of the binomial expansion of $(q+p)^n$. That is why it is called binomial distribution.

NOTE 3 ÷ n and p occuring in the binomial distribution are called the parameters of distribution.

NOTE 4 ÷ In a binomial distribution.

i) n, the number of trials is finite

ii) each trials has only two possible outcomes usually called success and failures

iii) all the trials are independent

iv) p (and hence q) is constant for all that trials.

Recurrence / Recursion Formula for Binomial Distribution.

$$P(r) = nC_r \, q^{n-r} p^r = \frac{n!}{(n-r)!\,r!} q^{n-r} p^r.$$

$$P(r+1) = nC_{r+1} \, q^{n-r-1} p^{r+1} = \frac{n!}{(n-r-1)!\,(r+1)!} q^{n-r-1} p^{r+1}.$$

$$\therefore \frac{P(r+1)}{P(r)} = \frac{(n-r)!}{(n-r-1)!} \times \frac{r!}{(r+1)!} \times \frac{p}{q}$$

$$= \frac{(n-r)\,(n-r-1)!}{(n-r-1)!} \times \frac{r!}{(r+1)\,r!} \, \frac{p}{q} = \left(\frac{n-r}{r+1}\right) \frac{p}{q}$$

$$\Rightarrow P(r+1) = \frac{n-r}{r+1} \, \frac{p}{q} \, P(r).$$

Which is the required recurrence formula. Applying this formula successively, we can find $P(1)$, $P(2)$, $P(3)$ ... if $P(0)$ is known.

Properties

1. This is distribution of discrede variable.
2. n and p are parameters of B.D
3. Mean of BD and np which shows the avg no. of success
4. variance of BD is npq And $SD = \sqrt{npq}$
5. When p and q are equal ie $p=q=\frac{1}{2}$ BD is a symmetrical distribution
6. When $p < \frac{1}{2}$ is skewness is positive and when $p > \frac{1}{2}$ it's skewness is negative
7. variance is always less than it's mean.
8. When n is very large and p or q is very small, Binomial distribution tend to poisson distribution
9. when n is very large and p or q are not very small, binomial distribution tend to Normal distribution.

Applications.

1. A manufacturing plant labels items as either defective or acceptable
2. A firm bidding for contracts will either get a contract or not
3. A marketing research firm receives survey responses of "Yes I will by" or "No I will not"
4. New job applicants either accept the offer or reject it.

Poison Distribution

If the parameters n and p of a binomial distribution are known, we can find the distribution. But in situations where n is very large and p is very small, application of binomial distribution is very laboumous. However, if we assume that as $n \to \infty$ and $p \to 0$ such that np always remains infinite, say m, we get the poison approximation to the binomial distribution.

Now for a binomial distribution

$$P(X=r) = {}^nC_r \, q^{n-r} \, p^r$$

$$= \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} (1-p)^{n-r} \, p^r$$

$$= \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} \left(1-\frac{m}{n}\right)^{n-r} \left(\frac{m}{n}\right)^r$$

since $np = m$ $\therefore p = \frac{m}{n}$

$$= \frac{m^r}{r!} \frac{n(n-1)(n-2)\cdots(n-r+1)}{n^r} \frac{\left(1-\frac{m}{n}\right)^n}{\left(1-\frac{m}{n}\right)^r}$$

$$= \frac{m^r}{r!} \left(\frac{n}{n}\right)\left(\frac{n-1}{n}\right)\left(\frac{n-2}{n}\right)\cdots\left(\frac{n-r+1}{n}\right) \cdot \frac{\left(1-\frac{m}{n}\right)^n}{\left(1-\frac{m}{n}\right)^r}$$

$$= \frac{m^r}{r!} \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\cdots\left(1-\frac{r-1}{n}\right) \frac{\left[\left(1-\frac{m}{n}\right)^{-\frac{n}{m}}\right]^{-m}}{\left(1-\frac{m}{n}\right)^r}$$

Also $n \to \infty$, each of the $(r-1)$ factor

$\left(1-\frac{1}{n}\right), \left(1-\frac{2}{n}\right), \ldots, \left(1-\frac{r-1}{n}\right)$ tends to 1. Also $\left(1-\frac{m}{n}\right)^r$ tends to 1

since $\underset{x \to \pm\infty}{Lt} \left(1+\frac{1}{x}\right)^x = e$ the Napellain base $\therefore \left[\left(1-\frac{m}{n}\right)^{-\frac{n}{m}}\right]^{-m} \to e^{-m}$ as $n \to \infty$

Hence m the limiting case when $n \to \infty$ we have.

$$P(X=r) = \frac{m^r e^{-m}}{r!} \quad (r = 0, 1, 2, 3, \ldots) \qquad \text{— A}$$

Scanned by CamScanner

Where m is a finite number $= np$.

(A) represents probability distribution which is called the poison probability distribution.

NOTE 1 :- m is called parameter of distribution

NOTE 2 :- $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots$ to ∞

NOTE 3 :- The sum of probabilities $P(r)$ for $r = 0, 1, 2, 3, \cdots$ is 1 since

$$P(0) + P(1) + P(2) + P(3) + \cdots = e^{-\lambda} + \frac{me^{-m}}{1!} + \frac{m^2e^{-m}}{2!} + \frac{m^3e^{-m}}{3!} + \cdots$$

$$= e^{-m}\left(1 + \frac{m}{1!} + \frac{m^2}{2!} + \frac{m^3}{3!} + \cdots\right)$$

$$= e^{-m} \cdot e^{m}$$

$$= \underline{\underline{1}}$$

Recurrence Formula For The Poisson Distribution

For passon distribution, $P(r) = \frac{m^r e^{-m}}{r!}$ & $P(r+1) = \frac{m^{r+1} e^{-m}}{(r+1)!}$

$$\therefore \quad \frac{P(r+1)}{P(r)} = \frac{m\, r!}{(r+1)!} = \frac{m}{r+1} \quad \text{or} \quad P(r+1) = \frac{m}{r+1} P(r), \; r = 0, 1, 2, 3, \ldots$$

This is called Recurrence Formula For Poisson distribution

Properties.

1. Poison dicribution is a distribution of discreate random variable.

2. In poisson distribution mean = variance = m. Hence its standard deviation is $\sqrt{m}$. This is the acid test to be applied to any data which might appear to conform poisson distribution

3. The sum of any finite of independent poison variates is itself number a poisson variate, with mean esual to the sum of the means of those variates taken seperately.

Applications.

1. The count of α particles emitted per unit of time is usefull in analysis of any radio active substance.

2. Number of telephone calls recieved at a given switch board per small unit of time.

3. Number of deaths per day or week due to a rate decrease in big hospital

4. In industrial production to find the proportion of defects per unit length, per unit area etc.

5. The count of bacteria per cc in blood

6. Distribution of number of mis prints per page of book.

Normal Distribution

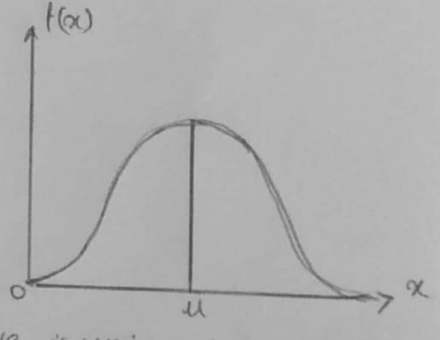
The normal distribution is a continous distribution. It can be derived from the binomial distribution in the limiting case when n, the number of trials is very large and p, the probability of a success, is close to $\frac{1}{2}$. The general equation of normal distribution is given by.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Where the variable x can assume all values from $-\infty$ to ∞, μ and σ called the parameters of the distribution, are respectively the mean and the standard deviation of the distribution and $-\infty < \mu < \infty$, $\sigma > 0$. x is called the normal variate and $f(x)$ is called probability density function of normal distribution.

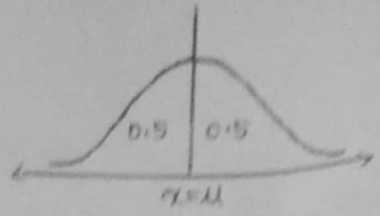
If a variable x has the normal distribution with mean μ and standard deviation σ, we briefly write $x : N(\mu, \sigma^2)$.

The graph of the normal distribution is called normal curve. It is bell-shaped and symmetrical about the mean μ. The two tails of the curve extend to ∞ to $-\infty$ towards the +ve and -ve directions of the x axis without ever metting it.



The curve is unimodal and the mode of normal distribution normal distribution coincides with it's mean μ. The line $x = \mu$ divides the area under the normal curve above x-axis into two equal parts. Thus the median of the distribution also coincides with its mean and mode. The area under the normal curve between any two given ordinates $x = x_1$ and $x = x_2$ represents the probability of values falling into the given interval. The total area under the normal curve above the x-axis is 1.

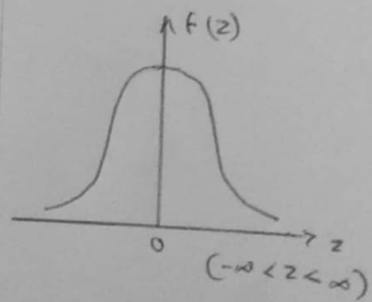
Properties.



1. Normal distribution is a continous distribution.
2. The mean of the normal distribution is μ and the variance is σ^2, standard deviation is σ.
3. Normal curve is bell shaped and it is symmetrical about the line $x=\mu$
4. $\int_{-\infty}^{\infty} f(x)\,dx = 1$ i.e total area under the normal curve is unity.
5. Since $f(x) \geq 0$ the curve always lies above the x-axis
6. The curve contains the maximum probability at $x=\mu$ and $f(x)_{max} = \frac{1}{\sqrt{2\pi}\sigma}$
7. As σ increases, the curve flattense out and as σ decreases, f increases.
8. $P(a \leq x \leq b) = \int_{a}^{b} f(x).dx$.
9. It is a unimodal distribution. The mean, mode and median concide.

Standard Form Of The Normal Distribution.

If X is a normal random variable with mean μ and standard deviation σ, then the random variable $z = \frac{X-\mu}{\sigma}$ has the normal distribution with mean 0 and standard deviation 1. The random variable z is called the standardized normal random variable.



The probability density function for the normal distribution in standard form is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

It is free from any parameter. This helps us to complete area under the normal probability curve by making use of standard tables.

NOTE 1 :- if $f(z)$ is the probability density function for the normal distribution then

$$P(z_1 \leq z \leq z_2) = \int_{z_1}^{z_2} f(z)\,dz = F(z_2) - F(z_1) \text{ where}$$

$$F(z) = \int_{-\infty}^{z} f(z)\,dz = P(z \leq z)$$

The function $F(z)$ defined above the distribution function for the normal distribution.

NOTE 2 :- The probabilities $P(z_1 \leq z \leq z_2)$, $P(z_1 < z \leq z_2)$, $P(z_1 \leq z < z_2)$ and $P(z_1 < z < z_1)$ are all regarded to be the same.

NOTE 3 :- $F(-z_1) = 1 - F(z_1)$.

Applications.

1. It's application goes beyond describing distributions
2. It is used by researchers and modelers.
3. The major use of normal distribution is the role it plays in statistical inference.
4. The z score along with the t-scene, Chi-squre and F statistics is important in hypothesis testing.
5. It helps managers / management to make decisions.

2. Derive the mean and variable of Binomial and Poisson distribution

ANSWER

a) Binomial Distribution

For the binomial distribution $P(x) = nC_x\, q^{n-x} p^x$

Mean $\mu = E(x)$

Mean $\mu = \sum_{r=0}^{n} x\, P(x) = \sum_{r=0}^{n} x \cdot nC_r\, q^{n-x} p^x$

$= 0 + 1 \cdot nC_1\, q^{n-1} p + 2 \cdot nC_2\, q^{n-2} p^2 + 3 \cdot nC_3\, q^{n-3} p^3 + \dots + n\, {}^nC_n\, p^n$

$= n q^{n-1} p + 2 \cdot \frac{n(n-1)}{2 \cdot 1} q^{n-2} p^2 + 3 \frac{n(n-1)(n-2)}{3 \cdot 2 \cdot 1} q^{n-3} p^3 + \dots + np^n$

$= npq^{n-1} + n(n-1) q^{n-2} p^2 + \frac{n(n-1)(n-2)}{2 \cdot 1} q^{n-3} p^3 + \dots + np^n$

$= np \left[q^{n-1} + (n-1) q^{n-2} p + \frac{(n-1)(n-2)}{2 \cdot 1} q^{n-3} p^2 + \dots + p^{n-1} \right]$

$= np \left[{}^{n-1}C_0\, q^{n-1} + {}^{n-1}C_1\, q^{n-2} + {}^{n-1}C_2\, q^{n-3} p^2 + \dots + {}^{n-1}C_{n-1}\, p^{n-1} \right]$

$= np(q+p)^{n-1} = \underline{np}$ $\quad (\because p+q=1)$

Hence mean of binomial distribution is $\underline{np}$.

Variance $\sigma^2 = E(x^2) - [E(x)]^2$

Variance $\sigma^2 = \sum_{r=0}^{n} x^2 P(x) - \mu^2 = \sum_{r=0}^{n} [x + x(x-1)] P(x) - \mu^2.$

$= \sum_{x=0}^{n} x P(x) + \sum_{r=0}^{n} x(x-1) P(x) - \mu^2 = \mu + \sum_{r=2}^{n} x(x-1)\, {}^nC_x q^{n-x} p^x - \mu^2$

(Since the contribution due to $x=0$ and $x=1$ is zero).

$= \mu + \left[2 \cdot 1 \cdot {}^nC_2\, q^{n-2} p^2 + 3 \cdot 2 \cdot {}^nC_3\, q^{n-3} p^3 + \dots + n(n-1)\, {}^nC_n p^n \right] - \mu^2$

$= \mu + \left[2 \cdot 1 \frac{n(n-1)}{2 \cdot 1} q^{n-2} p^2 + 3 \cdot 2 \frac{n(n-1)(n-2)}{3 \cdot 2 \cdot 1} q^{n-3} p^3 + \dots + n(n-1) p^n \right] - \mu^2$

Scanned by CamScanner

9.

$= \mu + [n(n-1)\, q^{n-2} p^2 + n(n-1)(n-2)\, q^{n-3} p^3 + \dots + n(n-1) p^n] - \mu^2$

$= \mu + n(n-1)\, p^2 [q^{n-2} + (n-2)\, q^{n-3} p + \dots + p^{n-2}] - \mu^2$

$= \mu + n(n-1) p^2 [{}^{n-2}C_0\, q^{n-2} + {}^{n-2}C_1\, q^{n-3} p + \dots + {}^{n-2}C_{n-2}\, p^{n-2}] - \mu^2$

$= \mu + n(n-1) p^2 (q+p)^{n-2} - \mu^2 = \mu + n(n-1) p^2 - \mu^2.$

$= np + n(n-1)p^2 - n^2p^2$ $\quad \because q+p=1$

$= np[1 + (n-1)p - np] = np[1-p] = \underline{npq}$ $\quad \because \mu = np.$

Hence the variance of the binomial distribution is npq.

Standard deviation of binomial distribution is $\sqrt{npq}$

b) Poisson Distribution.

For poisson distribution $P(r) = \dfrac{m^r e^{-m}}{r!}$

Mean $\mu = \sum_{r=0}^{\infty} r P(r) = \sum_{r=0}^{\infty} r \dfrac{m^r e^{-m}}{r!}$

$= e^{-m} \sum_{r=1}^{\infty} = e^{-m}\left(m + \dfrac{m^2}{1!} + \dfrac{m^3}{2!} + \dots\right)$

$= me^{-m}\left(1 + \dfrac{m}{1!} + \dfrac{m^2}{2!} + \dots\right) me^{-m} \cdot e^m = m.$

Thus the mean of poissons distribution is equal to the parameter $\underline{m}$.

variance $\sigma^2 = \sum_{r=0}^{\infty} r^2 P(r) - \mu^2 = \sum_{r=0}^{\infty} r^2 \dfrac{m^r e^{-m}}{r!} - m^2 = e^{-m} \sum_{r=1}^{\infty} \dfrac{r^2 m^r}{r!} - m^2$

$= e^{-m}\left[\dfrac{1^2 \cdot m}{1!} + \dfrac{2^2 m^2}{2!} + \dfrac{3^2 m^3}{3!} + \dfrac{4^2 m^4}{4!} + \dots\right] - m^2$

$= me^{-m}\left[1 + \dfrac{2m}{1!} + \dfrac{3m^2}{2!} + \dfrac{4m^3}{3!}\right] - m^2$

$= me^{-m}\left[1 + \dfrac{(1+1)m}{1!} + \dfrac{(1+2)m^2}{2!} + \dfrac{(1+3)m^3}{3!} + \dots\right] - m^2$

$= me^{-m}\left[e^m + m\left(1 + \dfrac{m}{1!} + \dfrac{m^2}{2!} + \dots\right)\right] - m^2$

$= me^{-m}[e^m + me^m] - m^2 = me^{-m} \cdot e^m(1+m) - m^2 = m(1+m) - m^2 = m$

Hence variance of Poisson distribution is also $\underline{m}$